No-three-in-line configurations with exact half-turn symmetry for $n = 36$, 37, 39, and a balance lemma for symmetric defects

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Abstract

We report $2n$-point no-three-in-line configurations on the $n \times n$ grid whose symmetry group is exactly the half-turn $\{1, \rho^2\}$, for $n = 36$ (two inequivalent configurations), 37 (two) and 39. For these sizes Flammenkamp's database (state 2026-08-11) listed no configuration in that class; a further such configuration is given for $n = 33$. The configurations were found by *exhaustive* branch-and-bound searches over small structured sub-classes of the half-turn class: quarter-turn-symmetric configurations with a controlled "defect" consisting of a few half-turn pairs. A simple *balance lemma* restricts the admissible defects to Eulerian digraphs on the row classes $\{i, n-1-i\}$; the smallest defects are one diagonal loop (Flammenkamp's pseudo-class rct4), a directed 3-cycle of pairs (odd n) and one loop on each long diagonal (even n). We give exhaustive counts of these families for small n, describe the solver and its validation against the known counts (OEIS A000769 and the class counts of the database), report measured growth rates of the search cost, and record negative results (empty families, failure of warm starts). Code, logs and configurations are available.

1 Introduction

The no-three-in-line problem (Dudeney, 1917) asks for the largest set of points of the $n \times n$ grid $\{0, \ldots, n-1\}^2$ containing no three collinear points. Since a line parallel to an axis contains at most two points, at most $2n$ points can be placed, and $2n$-point configurations are the objects of interest. Their existence for every n is a well-known open problem; a probabilistic heuristic (Guy–Kelly, corrected form) suggests that for large n fewer than $2n$ points can be placed. Computationally, $2n$-point configurations were known for all $n \leq 46$ and for $n = 48, 50, 52$ for decades [1, 2]; in 2025–2026 constraint programming [4], GPU branch-and-bound and SAT solving extended this to all $n \leq 70$ and to $n = 72, 74, 76$ (see the database [3], which also records, for each n, the number of configurations by their exact symmetry class under the dihedral group D_4 of the square, and marks the cells for which no configuration is known).

This note concerns those cells rather than the record sizes. Flammenkamp's table lists, for each n, the number of inequivalent $2n$-point configurations whose stabilizer in D_4 is exactly a given subgroup: trivial (*iden*), the half-turn (*rot2*), one diagonal reflection (*dia1*), one axis reflection (*ort1*), the quarter turns (*rot4*), both diagonal reflections (*dia2*), both axis reflections (*ort2*) and the full group (*full*), plus the pseudo-class *rct4* $\subset$ rot2 ("rot4 symmetry except on the long diagonals"). As of 2026-08-11 the rot2 column was complete for $n \leq 31$ and empty (no configuration known) for $n = 36, 37, 39, 41, 45, 47, \ldots$.

Results. (i) A balance lemma (Section 2) describing which "defects" a half-turn-symmetric configuration may have relative to a larger subgroup $H \leq D_4$: the defect pairs must form

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an Eulerian digraph on the row classes. In particular a single half-turn pair must lie on a long diagonal (which explains why rct4 is the only one-pair pseudo-class), and the smallest larger defects are a directed 3-cycle of pairs (odd n) and one loop on each diagonal (even n). (ii) Exhaustive enumerations of these families for small n (Section 4); several of them are non-empty exactly where the pure classes are rare. (iii) The first known configurations with exact rot2 symmetry for $n = 36$ (two-loop family, two inequivalent configurations), $n = 37$ (3-cycle family, two inequivalent configurations) and $n = 39$ (3-cycle family), and a further one for $n = 33$; the configurations are listed in the Appendix and were verified twice by independent programs. (iv) A description of the exact solver and its validation, growth measurements for the cost of finding one $2n$-point configuration in the classes rot4, rct4 and rot2, and a list of negative results.

Related work. The best known lower bound, $3n/2 - o(n)$ points, is the modular-hyperbola construction of Hall, Jackson, Sudbery and Wild [6]; it has not been improved since 1975 (see the discussions in [8, 9]). All linear-size constructions are algebraic modulo a prime, and modulo a prime a set with no three collinear is an arc of the affine plane, of size at most $p + 1$ (Bose's bound), the arcs of that size being conics by Segre's theorem [7]; Green asked whether every large no-three-in-line set reduces to an algebraic curve modulo some prime (as quoted in [9]). For $k \geq 3$ points per line the asymptotic answer kn was obtained recently by probabilistic methods [8, 9], which explicitly do not apply to $k = 2$. The probabilistic heuristic of Guy and Kelly [5], in its corrected form [10, 4], predicts that $2n$-point configurations become rare for large n; Prellberg notes that the same heuristic wrongly predicts fewer than one symmetric configuration [4], and Flammenkamp explains the abundance of rotationally symmetric configurations by the fact that a pair of points blocks only $O(\log n)$ cells in the rotational classes against $\Theta(n)$ in the axis-reflection classes [3]. Machine-learning approaches (PatternBoost, reinforcement learning) have been compared with integer programming on this problem and do not scale beyond $n \approx 14$ [13]. We are not aware of an earlier formulation of the balance lemma or of the defect families, nor of previously published $2n$-point configurations with exact half-turn symmetry for $n = 36, 37, 39$.

2 Symmetry classes and the balance lemma

Let $m = n - 1$ and let D_4 act on the grid; write $\rho(u, v) = (v, m - u)$ for the quarter turn, so $\rho^2(u, v) = (m - u, m - v)$, $\sigma(u, v) = (v, u)$ for the reflection in the main diagonal and $\sigma'(u, v) = (m - v, m - u)$ for the reflection in the anti-diagonal. A $2n$-point no-three-in-line configuration has exactly two points in every row and every column.

Fix a subgroup $H \leq D_4$ containing ρ^2 and consider a rot2-symmetric configuration C (i.e. $\rho^2 C = C$) which is a disjoint union of H-orbits and of a set D of half-turn pairs $\{p, \rho^2 p\}$ (the *defect*) that are not H-orbits. For $a \in \{0, \ldots, m\}$ let $r_X(a)$, $c_X(a)$ denote the numbers of points of a set X in row a and in column a.

Lemma 1 (balance). *If H contains a motion mapping column a onto row a for every a (any quarter turn or any diagonal reflection does), then every H-orbit O satisfies $r_O(a) = c_O(a)$ for all a. Consequently the defect of a $2n$-point configuration satisfies $r_D(a) = c_D(a)$ for all a.*

Proof. If $g \in H$ maps column a onto row a then g maps $O \cap \{\text{column } a\}$ bijectively onto $O \cap \{\text{row } a\}$ because $gO = O$; hence $r_O(a) = c_O(a)$. Summing over the orbits and using $r_C(a) = c_C(a) = 2$ gives $r_D(a) = c_D(a)$. □

Interpret the pairs of D as arcs of a digraph on the *row classes* $\{i, m-i\}$: the pair $\{(x, y), (m-x, m - y)\}$ occupies rows $x, m - x$ and columns $y, m - y$, i.e. it is an arc from class $\bar{x}$ to class $\bar{y}$ (a loop if $\bar{x} = \bar{y}$, i.e. if the pair lies on a long diagonal). Lemma 1 says that in-degree equals out-degree at every class:

Corollary 2. *The defect D is an Eulerian digraph on the row classes, hence a union of directed cycles. A single defect pair must be a loop, i.e. must lie on a long diagonal. For $H = \langle\rho\rangle$ every usable H-orbit has four points (the centre cell, a fixed point for odd n, cannot occur), so $2n = 4a + 2|D|$ and $|D| \equiv n \pmod 2$: for odd n the smallest defects are one loop — this is exactly the pseudo-class rct4 of [3] — and a directed 3-cycle $(x, y), (y, z), (z, x')$ with $x' \in \{x, m - x\}$; for even n the smallest defects are two loops. (For $H = \langle\sigma, \sigma'\rangle$ the diagonal 2-orbits $\{(i, i), (m - i, m - i)\}$ are themselves H-orbits, and the parity condition counts them together with D.)*

For $H = \langle\rho\rangle$ (rot4) an H-orbit off the diagonals is $\{(u, v), (v, m-u), (m-u, m-v), (m-v, u)\}$; for $H = \langle\sigma, \sigma'\rangle$ (dia2) it is $\{(u, v), (v, u), (m - u, m - v), (m - v, m - u)\}$. We call these the C4 base and the V2 base. The following families are therefore the natural next steps beyond rct4:

- **3-cycle family** (odd n): $(n-3)/2$ orbits of the C4 base (or of the V2 base) plus a directed 3-cycle of half-turn pairs;
- **two-loop family** (even n): $n/2 - 1$ orbits of the C4 base plus one loop on each long diagonal, $\{(x, x), (m - x, m - x)\}$ and $\{(y, m - y), (m - y, y)\}$ with $y \notin \{x, m - x\}$ (for $y \in \{x, m - x\}$ the two loops form a rot4-orbit and the configuration is rot4-symmetric).

A configuration of these families is not H-symmetric (the defect breaks H), so its stabilizer is a proper subgroup of H containing ρ^2; in every case found below the stabilizer is exactly $\{1, \rho^2\}$ (verified explicitly), so the configurations fall into the rot2 column of the database. A two-loop configuration is "rot4 symmetric except on the long diagonals", i.e. of rct4 type for even n (the database keeps rct4 as a pseudo-class for odd n only).

Remark 3. The families of "mixed bases" — C4-orbits together with at least one V2-orbit (plus one loop for odd n) — are admissible by Lemma 1 but turned out to be empty in all our enumerations (all $n \le 23$ exhaustively, and randomized searches up to $n = 33$); we have no structural explanation.

3 The solver

All searches were done with an exact branch-and-bound program (`there.c`, about 600 lines of C, no external libraries). Cells are $t = un + v$; a state keeps two bit-matrices, *alive* by rows and by columns. Every pair of chosen points "shades" all grid cells of the line through them (walked with the primitive direction obtained from a gcd table); shaded cells die, as do all cells of a row or column that already holds two points. A symmetry class is imposed by choosing whole orbits: a cell is a candidate only if its whole orbit is alive, and the "orbit-alive" bit-masks are derived from the alive matrices by transposition and bit reversal (one word operation per group element). Branching uses the most constrained reading class (row or column with the fewest orbit-alive cells) and dies as soon as some class cannot be completed. A failed candidate is killed for its siblings, so that a search with a fixed symmetry class is complete: "no configuration" is a proof of absence for that class. With restarts and randomized tie-breaking the same program is used to find one configuration quickly. A `PAIRS` option fixes a set of half-turn pairs (the defect), kills their images under the base group, and searches the remaining orbits of the base; the same option with a diagonal pair reproduces rct4 exactly.

Validation. For $n \le 10$ the program enumerates all $2n$-point configurations without symmetry: $1, 2, 11, 32, 50, 132, 380, 368, 1135$ labelled configurations for $n = 2, \ldots, 10$, which reduce to $1, 1, 4, 5, 11, 22, 57, 51, 156$ classes under D_4, in agreement with OEIS A000769. For symmetric classes we compared the number N_H of labelled configurations invariant under H with the class counts c_K of [3] through $N_H = \sum_{K \supseteq H} c_K \cdot |D_4|/|K|$: e.g. rot2 for $n = 8$: $36 = 4 \cdot 7 + 2 \cdot 4$;

$n = 10$: $67 = 4 \cdot 13 + 2 \cdot 6 + 2 \cdot 1 + 1$; dia1 for $n = 13$: $20 = 2 \cdot 9 + 2 \cdot 1$; $n = 15$: $28 = 2 \cdot 13 + 2 \cdot 1$; dia2 for $n = 11, \ldots, 25$: $0, 2, 2, 0, 0, 0, 2, 4 = 2 \cdot (0, 1, 1, 0, 0, 0, 1, 2)$; rct4 for $n = 9, 17, 19, 21$: $2, 2, 4, 2 = 2 \cdot (1, 1, 2, 1)$ — all in agreement. Two independent implementations (the two agents' programs) agree on every enumeration they both performed. Every configuration reported here was finally re-checked by an independent script that tests all $\binom{2n}{3}$ triples in exact integer arithmetic and computes the stabilizer explicitly.

4 Results

4.1 Exhaustive enumerations of the defect families

The defect sets were enumerated up to D_4 (one representative per orbit of D_4 acting on defects); for each representative the sub-class was searched exhaustively. Table 1 gives, for odd n, the number of inequivalent 3-cycle defects and the number of pairwise inequivalent configurations found for the two bases; Table 2 gives the two-loop family for even n. Counts marked with $\geq$ come from sweeps that were still running when this draft was written (the number of sub-classes already swept is stated); the entries will be finalized. (Sub-classes whose fixed defect cells already violate the law — e.g. 24 of the 2240 at $n = 33$, 28 of 3264 at $n = 37$ — are trivially empty and are counted as swept.)

n	3-cycle defects (mod D_4)	C4 base	V2 base	status of the sweep
13	80	5	2	complete
15	140	0	0	complete
17	224	0	1	complete
19	336	0	0	complete
21	480	0	0	complete
23	660	0	0	complete
25	880	0	0	complete
33	2240	1	0	complete (both bases)
37	3264	2	0	complete (both bases)
39	3876	≥ 1	≥ 0	C4 base: 300 of 3876 sub-classes swept (1 configuration); V2 base: 3277 of 3876 swept (none)
41	4560	≥ 0	0	V2 base complete; C4 base: 3030 of 4560 sub-classes swept (none)
45	6160	—	≥ 0	V2 base: 400 of 6160 sub-classes swept (none); C4 base not started

Table 1: The 3-cycle family (odd n): C4- or V2-orbits plus a directed 3-cycle of half-turn pairs; numbers of pairwise inequivalent configurations. A sub-class is one defect (one directed 3-cycle up to D_4) together with one base; "complete" means every sub-class was searched exhaustively.

4.2 First configurations with exact half-turn symmetry for $n = 36, 37, 39$

Table 3 lists the new configurations; their coordinates and their encodings in the 90-character format of [3] are given in the Appendix. "First known" refers to the state of the database on 2026-08-11, in which the rot2 cells for $n = 36, 37, 39$ were empty; the $n = 33$ configuration is new but the cell was not empty.

4.3 Cost of finding one configuration, and growth

For orientation we report measured costs of finding one $2n$-point configuration on the $n \times n$ grid with the solver of Section 3, on two consumer machines (Apple M3 laptop, 8 cores; a 12-core

n	sub-classes (x,y)	inequivalent configurations
14–22	all	0
24	132	2
26	156	1
28	182	1
36	306	≥ 2 — finding mode (about 2 min per sub-class): 81 sub-classes tried, two configurations found in two of them, one sub-class exhausted empty, the rest timed out; an exhaustive sweep is in progress

Table 2: The two-loop family (even n): C4-orbits plus one half-turn loop on each long diagonal. Sub-classes are the ordered pairs (x,y), $0 \leq x, y < n/2$, $x \neq y$ (x = row class of the main-diagonal loop, y = row class of the anti-diagonal loop); the quarter turn maps the sub-class (x,y) onto (y,x), so $n(n-2)/8$ of them suffice for the enumeration. For $n \leq 28$ every sub-class was searched exhaustively.

n	family and defect	remark
36	two loops: $(6,6),(29,29)$ and $(5,30),(30,5)$	first known exact rot2
36	two loops: $(1,1),(34,34)$ and $(9,26),(26,9)$	second, inequivalent (different loop rows)
37	3-cycle $(1,3),(3,31),(31,1)$, C4 base	first known exact rot2
37	3-cycle $(2,4),(4,20),(20,34)$, C4 base	second, inequivalent
39	3-cycle $(4,8),(8,20),(20,34)$, C4 base	first known exact rot2
33	3-cycle $(2,3),(3,19),(19,2)$, C4 base	new (cell not empty)

Table 3: New $2n$-point configurations with stabilizer exactly $\{1, \rho^2\}$.

x86 server; 8–12 seeds in parallel; wall time to the first configuration): rot4 (even n): $n = 36$: 10–60 s, $n = 40$: 70 s; rct4 (odd n): $n = 41$: 806 s, $n = 43$: about 6.7 core-hours, $n = 45$: none within 3.3 core-hours; rot2: $n = 24$: 10 s, $n = 27$: 137 s. The cost grows by roughly a factor 8–10 per +4 in n for rot4/rct4 and by a factor of about 6 per +2 for rot2, consistent with the exhaustive tree sizes (rot4: $\times 20$ per +4; rot2 and dia1: $\times$9–10 per +2) divided by the slowly growing numbers of configurations, and consistent with the growth reported for CP-SAT in [4]. Extrapolated to the open sizes $71, 73, 75, 77, \ldots$ this gives 10^5–10^7 core-hours for our program; the record computations of 2026 (SAT and GPU search on clusters) are of that order or use better tools. Sub-class sweeps of the defect families are cheap because a fixed defect shades a large part of the grid: a sub-class of the 3-cycle family at $n = 37$–41 is exhausted in seconds to a few minutes.

4.4 Negative results

- A single defect pair off the diagonals is impossible (Corollary 2); the "rot4 except one pair anywhere" family is empty, as our enumerations for $n \leq 27$ had shown before the lemma was found.

- Mixed bases (C4-orbits with at least one V2-orbit, plus a loop for odd n) are empty for all $n \leq 23$ (exhaustive) and no configuration was found by randomized search for $n \leq 33$.

- Warm starts do not work: shifting a known rot4 configuration for $n = 44$ into the 46×46 grid (which preserves all its collinearities) and freeing k of its 22 orbits, no completion exists even for $k = 15$ (exhaustive); configurations of neighbouring sizes appear to be uncorrelated.

- The known extremal configurations are not "algebraic modulo a prime" in any cheap sense:

for the rot4/rct4 configurations of the database with $41 \leq n \leq 76$ (142 configurations) the largest subset lying on one line, axis-parallel parabola or shifted hyperbola over $\mathbb{F}_p$, maximised over primes $11 \leq p \leq 61$, is on average 16% of the points (never above 22%), *less* than for random sets with two points per row (19% on average). In the finite range the $2n$-point sets do not reduce modulo a prime to a low-degree curve (cf. Green's remark on Problem 72 of [11]), while the best asymptotic constructions do.

- On the "algebraic" side, the modular-hyperbola construction of [6] is optimal within its own $2p \times 2p$ box: no lawful subset of the hyperbola in that box has more than $3(p-1)$ points, and the maximum sets are classified; unions of two hyperbolae in the box gain only a bounded amount as far as computed ($+5, +5, +6, +5$ over $3(p-1)$ for $p = 11, 13, 17, 19$, exact; at $p = 23$ between $+4$ and $+12$). These results are proved and documented in the companion note [12] from the same project.
- Local search on exactly $2n$ points (min-conflicts with tabu, simulated annealing) fails from $n \approx 12$–28 on, although it produces large partial configurations easily (e.g. 134 points on the 71×71 grid); belief-propagation decimation without backtracking fails from $n = 8$ on.

5 Reproducibility

The programs, the sub-class lists, the sweep logs and the certified configurations are in the public repository `https://github.com/iwasborninbali/saturation` (immutable tag `defects-v0.6`; checksums in `docs/RELEASES.md`): `there.c`; `kit71/there_twisted_experiment.c` — a variant of `there.c` with the two bases and the `PAIRS` defect built in, used for the family sweeps and driven by `kit71/bench/family_sweep.sh` over the canonical defect lists produced by `kit71/bench/cycle_family.py`; `web/stab.py` for the exact class and the database encoding; `docs/SUBMISSION.md` for the configurations. Typical commands: `ALL=1 PAIRS="1,3;3,31;31,1" ./there 37 2` enumerates the sub-class that contains the first $n = 37$ configuration (one configuration, $1.4 \cdot 10^7$ nodes); `PAIRS="6,6;5,30" ./there 36 2` finds the first $n = 36$ configuration in about 100 s and `PAIRS="1,1;9,26 ./there 36 2` the second in about 30 s (wall time on the laptop); the sub-class $(6, 5)$ is exhausted in about 7 min and contains exactly two labelled configurations, i.e. one up to the diagonal reflections, which preserve both loops.

6 AI/tool-use disclosure and authorship statement

Following the arXiv policy on generative AI, the AI systems involved are not listed as authors; their role is disclosed here. This work was carried out by two autonomous AI agents (Claude Fable 5, Anthropic), running as independent solvers on two machines and communicating through the repository, under the direction of the author of record (a third agent of the same kind joined the project later and worked on the companion note [12]). The agents wrote the programs, designed and ran the searches, found and proved the balance lemma (independently of each other, in two formulations), verified each other's results, and drafted this text; the author of record posed the problem, provided the computing environment, made all decisions about scope and publication, reproduced the computational checks, and takes responsibility for the content. We estimate that about nine tenths of the technical work was done by the AI systems; we state this plainly because it is true. All configurations reported here are verifiable by anyone with the scripts in the repository (every triple checked in exact integer arithmetic).

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A The configurations

Rows and columns are numbered $0, \dots, n-1$ from the top-left corner; the encoding is the class character followed by two column characters per row, with the 90-character alphabet of [3]:

`0-9A-Za-z#$%&@?!()[]<>{}=*+|-/~^_:;,.`

$n = 36$ **(configuration 1), 72 points, stabilizer** $\{1, \rho^2\}$**.** Encoding:
`:3NEHFSVZ3FPU6E2BNQ8I5GDS08JOTYVXDP9Y1QAM2416BGRZ7MJUHR9COXLT5AKW047KILCW`
Rows u : v_1, v_2 (row u has points at columns v_1, v_2):
0: 3,23 1: 14,17 2: 15,28 3: 31,35 4: 3,15 5: 25,30 6: 6,14 7: 2,11
8: 23,26 9: 8,18 10: 5,16 11: 13,28 12: 0,8 13: 19,24 14: 29,34 15: 31,33
16: 13,25 17: 9,34 18: 1,26 19: 10,22 20: 2,4 21: 1,6 22: 11,16 23: 27,35
24: 7,22 25: 19,30 26: 17,27 27: 9,12 28: 24,33 29: 21,29 30: 5,10 31: 20,32
32: 0,4 33: 7,20 34: 18,21 35: 12,32

$n = 36$ **(configuration 2), 72 points, stabilizer** $\{1, \rho^2\}$**.** Encoding:
`cJK1J9OCK6AORRVEH56QXLV25MWCIAS0301DS7MYZWZ7PHN3DUX4E29TUIL488BPTFNBQGYFG`
Rows u : v_1, v_2 (row u has points at columns v_1, v_2):
0: 19,20 1: 1,19 2: 9,24 3: 12,20 4: 6,10 5: 24,27 6: 27,31 7: 14,17
8: 5,6 9: 26,33 10: 21,31 11: 2,5 12: 22,32 13: 12,18 14: 10,28 15: 0,3
16: 0,1 17: 13,28 18: 7,22 19: 34,35 20: 32,35 21: 7,25 22: 17,23 23: 3,13
24: 30,33 25: 4,14 26: 2,9 27: 29,30 28: 18,21 29: 4,8 30: 8,11 31: 25,29
32: 15,23 33: 11,26 34: 16,34 35: 15,16

$n = 37$ **(configuration 1), 74 points, stabilizer** $\{1, \rho^2\}$**.** Encoding:
`:GJ39EP7VENDZJRSX7P6ZCF28IQ4VWYKQFa06COUa0LAG245WAISYLO1UBT389H1NDM5TBMRXHK`
Rows u : v_1, v_2 (row u has points at columns v_1, v_2):
0: 16,19 1: 3,9 2: 14,25 3: 7,31 4: 14,23 5: 13,35 6: 19,27 7: 28,33
8: 7,25 9: 6,35 10: 12,15 11: 2,8 12: 18,26 13: 4,31 14: 32,34 15: 20,26
16: 15,36 17: 0,6 18: 12,24 19: 30,36 20: 0,21 21: 10,16 22: 2,4 23: 5,32
24: 10,18 25: 28,34 26: 21,24 27: 1,30 28: 11,29 29: 3,8 30: 9,17 31: 1,23
32: 13,22 33: 5,29 34: 11,22 35: 27,33 36: 17,20

$n = 37$ **(configuration 2), 74 points, stabilizer** $\{1, \rho^2\}$**.** Encoding:
`:FJ7E4PBGKO6U5VDZDH9RCF2X4QSTIZQa2X0SEM8a3Y0A1I78AW3YLO9RJN1N5V6UCGKPBWMTHL`
Rows u : v_1, v_2 (row u has points at columns v_1, v_2):
0: 15,19 1: 7,14 2: 4,25 3: 11,16 4: 20,24 5: 6,30 6: 5,31 7: 13,35
8: 13,17 9: 9,27 10: 12,15 11: 2,33 12: 4,26 13: 28,29 14: 18,35 15: 26,36
16: 2,33 17: 0,28 18: 14,22 19: 8,36 20: 3,34 21: 0,10 22: 1,18 23: 7,8
24: 10,32 25: 3,34 26: 21,24 27: 9,27 28: 19,23 29: 1,23 30: 5,31 31: 6,30
32: 12,16 33: 20,25 34: 11,32 35: 22,29 36: 17,21

$n = 39$**, 78 points, stabilizer** $\{1, \rho^2\}$**.** Encoding:
`:CGAWLU7M8CAN1FPZ2K9TXbOPYc7BBL5W3c2E4JIKJYOa0Z6XHRRV04DE159TIa3DNbFSQUGV8H6SMQ`
Rows u : v_1, v_2 (row u has points at columns v_1, v_2):
0: 12,16 1: 10,32 2: 21,30 3: 7,22 4: 8,12 5: 10,23 6: 1,15 7: 25,35
8: 2,20 9: 9,29 10: 33,37 11: 24,25 12: 34,38 13: 7,11 14: 11,21 15: 5,32
16: 3,38 17: 2,14 18: 4,19 19: 18,20 20: 19,34 21: 24,36 22: 0,35 23: 6,33
24: 17,27 25: 27,31 26: 0,4 27: 13,14 28: 1,5 29: 9,29 30: 18,36 31: 3,13
32: 23,37 33: 15,28 34: 26,30 35: 16,31 36: 8,17 37: 6,28 38: 22,26

$n = 33$**, 66 points, stabilizer** $\{1, \rho^2\}$**.** Encoding:
`:MOAC3H9J6GJOBS9B05PT0VPQHV5UEI2C4SKUEI2R1F671W37RWLN4L8DGQDNFTKM8A`
Rows u : v_1, v_2 (row u has points at columns v_1, v_2):
0: 22,24 1: 10,12 2: 3,17 3: 9,19 4: 6,16 5: 19,24 6: 11,28 7: 9,11
8: 0,5 9: 25,29 10: 0,31 11: 25,26 12: 17,31 13: 5,30 14: 14,18 15: 2,12
16: 4,28 17: 20,30 18: 14,18 19: 2,27 20: 1,15 21: 6,7 22: 1,32 23: 3,7

24: 27,32 25: 21,23 26: 4,21 27: 8,13 28: 16,26 29: 13,23 30: 15,29 31: 20,22
32: 8,10